

PROJECT TITLE

SMART CITY TRAFFIC MANAGEMENT & NAVIGATION SYSTEM

Submitted in Partial Fulfillment of the Requirements for the Award of the Degree of

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in

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ABSTRACT

The rapid growth of urban populations and the increasing number of vehicles have created significant challenges in managing traffic efficiently. Congestion on roads not only increases travel time but also leads to higher fuel consumption, environmental pollution, and reduced productivity. Traditional navigation systems primarily focus on finding the shortest geographical route without considering real-time traffic conditions or road availability. As a result, users often experience delays despite selecting the shortest path.

The **Smart City Traffic Management & Navigation System** is designed to address these challenges by providing an intelligent route optimization solution. The system models the road network as a graph, where intersections are represented as nodes and roads as weighted edges. To determine the most efficient route between two locations, the system uses **Dijkstra's Shortest Path Algorithm**, which calculates the minimum-cost path based on both road distance and traffic conditions. The weight of each road dynamically changes according to traffic intensity, enabling the system to avoid congested routes and recommend faster alternatives.

The backend of the application is developed using **Node.js** and **Express.js**, while **MySQL** is used for secure storage and efficient management of user accounts, road information, junction data, and traffic details. RESTful APIs facilitate communication between the client interface and the backend services, ensuring smooth data exchange and system scalability.

This project demonstrates the practical application of graph algorithms in solving real-world transportation problems. It contributes to the development of smart city infrastructure by improving route planning, reducing travel time, lowering fuel consumption, and enhancing the overall efficiency of urban transportation systems. Furthermore, the system has the potential to be extended with technologies such as GPS, IoT sensors, Artificial Intelligence, and live traffic monitoring to provide a fully automated and intelligent navigation platform.

Keywords: Smart City, Traffic Management, Navigation System, Route Optimization, Graph Data Structure, Dijkstra Algorithm, Node.js, Express.js, MySQL, REST API.

CHAPTER 1

INTRODUCTION

1.1 Introduction

The rapid growth of urbanization and industrialization has significantly increased the number of vehicles on roads across the world. As cities continue to expand, managing traffic efficiently has become one of the biggest challenges faced by governments and transportation authorities. Heavy traffic congestion not only increases travel time but also causes fuel wastage, air pollution, road accidents, and economic losses. Therefore, there is a growing need for intelligent transportation systems that can optimize traffic flow and provide efficient navigation services.

A Smart City Traffic Management & Navigation System is an advanced solution designed to improve urban transportation by utilizing modern software technologies and efficient algorithms. Unlike traditional navigation systems that calculate routes based only on geographical distance, this system also considers real-time traffic conditions, road congestion, and road availability before recommending the best route. This intelligent approach enables users to reach their destinations faster while reducing unnecessary delays and fuel consumption.

The proposed system represents the city's transportation network as a graph data structure. In this graph, each road junction is represented as a node, while the roads connecting the junctions are represented as weighted edges. The weight assigned to each road depends on factors such as road distance and traffic conditions. By using graph theory and shortest path algorithms, the system can calculate the most efficient route between any two locations within the city.

To perform route optimization, the system implements Dijkstra's Shortest Path Algorithm. The algorithm efficiently calculates the minimum-cost path between a source and destination by continuously selecting the node with the lowest accumulated cost. Unlike conventional implementations of Dijkstra's Algorithm, this project introduces dynamic edge weights based on traffic levels such as Low, Medium, and High. This modification enables the algorithm to avoid heavily congested roads and recommend faster alternative routes.

The backend of the system is developed using Node.js and Express.js, providing a scalable and efficient server-side environment. MySQL is used as the relational database management system for storing user information, road details, junction information, and traffic data. RESTful APIs are implemented to establish communication between the frontend interface and backend services, ensuring secure and efficient data exchange.

Overall, the Smart City Traffic Management & Navigation System provides an intelligent platform for route optimization, traffic management, and transportation planning. It demonstrates how Data Structures and Algorithms can be applied to solve real-world transportation problems and lays the foundation for future smart city applications.

1.2 Background of the Project

Urban transportation has undergone significant changes over the last decade due to the rapid increase in population and vehicle ownership. According to recent transportation studies, traffic congestion is responsible for millions of hours of travel delay every year, resulting in increased fuel consumption and environmental pollution.

Most existing navigation applications focus mainly on identifying the shortest geographical distance between two locations. However, the shortest distance does not always guarantee the fastest journey because road conditions constantly change due to traffic congestion, road maintenance, accidents, and weather conditions. As a result, users may experience unnecessary delays despite following the recommended route.

The concept of smart cities encourages the integration of information technology, communication systems, and intelligent software to improve public services. Intelligent traffic management is one of the major components of a smart city because efficient transportation directly contributes to economic growth, environmental sustainability, and better quality of life.

The Smart City Traffic Management & Navigation System has been developed to overcome these limitations. Instead of considering only the physical distance, the system calculates an optimized route by incorporating traffic intensity into the route calculation process. This approach provides users with more realistic and efficient navigation recommendations.

1.3 Need for the Project

Modern cities require intelligent transportation systems capable of handling rapidly changing traffic conditions. The traditional approach of selecting the shortest route often fails because it ignores several practical factors that affect travel time.

The need for this project arises due to the following reasons:

- Rapid increase in urban traffic congestion.
- Longer travel times caused by inefficient route planning.
- High fuel consumption due to traffic delays.
- Increased air pollution caused by vehicles remaining idle in traffic.
- Lack of dynamic traffic-aware navigation in conventional systems.
- Requirement for smart transportation solutions in modern cities.
- Efficient utilization of existing road infrastructure.
- Better route planning for emergency services such as ambulances and fire departments.

The proposed system addresses these issues by implementing an intelligent graph-based navigation model that continuously calculates optimized routes according to current traffic conditions.

1.4 Project Overview

The Smart City Traffic Management & Navigation System is a web-based application developed to provide intelligent route optimization for urban transportation. The system models the complete road network as a graph where:

- Junctions are represented as graph nodes.
- Roads are represented as graph edges.
- Distance and traffic conditions are represented as edge weights.

Whenever a user enters the source and destination locations, the system retrieves the required road information from the database and constructs the corresponding graph. Dynamic weights are assigned to each road according to traffic intensity. Dijkstra's Shortest Path Algorithm is then applied to calculate the minimum-cost path. Finally, the optimized route is displayed to the user along with the estimated travel distance.

The application also provides administrative features such as managing road information, updating traffic conditions, maintaining junction records, and monitoring transportation data. This architecture makes the system suitable for both navigation services and traffic management applications.

1.5 Features of the System

The major features of the proposed system include:

- Intelligent route optimization using graph algorithms.
- Dynamic traffic-aware navigation.
- Efficient shortest path calculation using Dijkstra's Algorithm.
- User registration and authentication.
- Road and junction management.
- Traffic condition updates.
- RESTful API communication.
- Secure database management using MySQL.
- Scalable backend architecture using Node.js and Express.js.
- Future support for GPS, IoT devices, and Artificial Intelligence.

CHAPTER 2

ALGORITHM

2.1 Introduction

Algorithms are the backbone of every intelligent software system. In the **Smart City Traffic Management & Navigation System**, the algorithm is responsible for identifying the most efficient route between two locations by analyzing the entire road network. Since a city's transportation system consists of multiple roads and junctions, it can be represented using a **Graph Data Structure**. Each junction acts as a node, and every road connecting two junctions is represented as an edge with an associated weight.

The system uses **Dijkstra's Shortest Path Algorithm**, one of the most efficient algorithms for finding the shortest path in a weighted graph with non-negative edge weights. Unlike traditional navigation systems, this project enhances the algorithm by dynamically modifying edge weights according to traffic conditions. This enables the system to avoid congested roads and provide users with faster and more practical routes.

2.2 Graph Data Structure

A Graph is a non-linear data structure consisting of a collection of vertices (nodes) and edges (connections).

It is mathematically represented as:

$$G = (V, E)$$

Where:

- V = Set of Vertices (Junctions)
- E = Set of Edges (Roads)

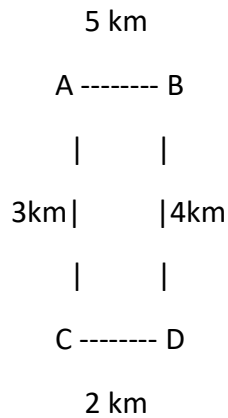
In this project:

Graph Component	Real-World Meaning
Vertex (Node)	Road Junction
Edge	Road
Weight	Distance + Traffic Cost

The graph-based representation makes route calculation highly efficient even when the road network becomes large.

2.3 Graph Representation

Consider the following road network:



The graph representation becomes:

$A \rightarrow B$ (5)

$A \rightarrow C$ (3)

$B \rightarrow D$ (4)

$C \rightarrow D$ (2)

Each road stores:

- Source Junction
- Destination Junction
- Distance
- Traffic Level
- Road Status

This information is retrieved from the MySQL database before route calculation begins.

2.4 Adjacency List Representation

Instead of storing the graph in matrix form, the system uses an **Adjacency List** because it requires less memory and provides faster traversal for sparse road networks.

Example:

1:

→ (2,5)

→ (3,3)

2:

→ (4,4)

3:

→ (4,2)

Advantages

- Memory Efficient
- Faster Graph Traversal
- Suitable for Large Cities
- Easy to Update Roads
- Supports Dynamic Traffic Updates

2.5 Dijkstra's Shortest Path Algorithm

Overview

Dijkstra's Algorithm is a greedy algorithm introduced by Edsger W. Dijkstra in 1956. It is used to calculate the shortest path between a source node and all other nodes in a graph with non-negative edge weights.

In this project, the algorithm calculates the minimum-cost route between the source and destination while considering road distance and traffic conditions.

2.6 Working Principle

The algorithm follows these steps:

Step 1

Initialize all nodes with infinite distance.

Source = 0

All Others = ∞

Step 2

Select the node having the minimum temporary distance.

Initially,

Source Node = 0

Step 3

Visit all neighbouring nodes.

Calculate

New Distance = Current Distance + Road Weight

If the new distance is smaller than the previous distance, update it.

Step 4

Mark the processed node as visited.

Visited nodes are not processed again.

Step 5

Repeat the process until

- Destination is reached
- or
- Every node has been visited.

Finally,

Display the optimized route.

2.7 Traffic-Based Weight Calculation

Traditional Dijkstra Algorithm uses only the road distance.

Our project improves it by introducing **dynamic traffic weights**.

Formula :

Final Weight = Distance × Traffic Factor

Traffic Factors:

Traffic Level	Factor
Low	1
Medium	1.5
High	3

Example

Road Distance

8 km

Traffic : HIGH

Calculation Weight= $8 \times 3 = 24$

Therefore,

Although the actual distance is 8 km, the algorithm considers its effective cost as 24, making congested roads less preferable during route selection.

2.8 Dijkstra Algorithm (Pseudocode)

Algorithm Dijkstra(Graph, Source)

1. Initialize all distances as Infinity
2. Set Source Distance = 0
3. Insert Source into Priority Queue
4. While Queue is not Empty

 Select Node having Minimum Distance

 For each Neighbour

 Calculate New Distance

 If New Distance < Current Distance

 Update Distance

Update Parent

5. Return Shortest Path

2.9 Complexity Analysis

Time Complexity

Using Priority Queue

$$O((V + E) \log V)$$

Where

- V = Number of Vertices
- E = Number of Edges

Space Complexity

$$O(V + E)$$

This complexity makes Dijkstra's Algorithm suitable for medium and large transportation networks.

2.10 Route Optimization Workflow

The complete route calculation process is shown below.

User Selects

Source & Destination

|



Fetch Road Data from MySQL Database

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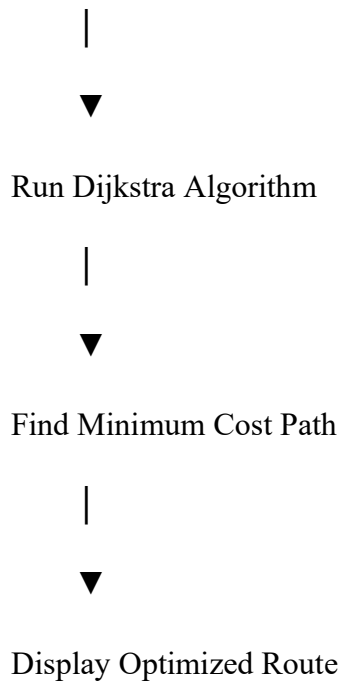


Create Graph Representation

|



Apply Traffic Weight Calculation



2.11 Advantages of Using Dijkstra Algorithm

The implementation of Dijkstra's Algorithm provides several benefits to the Smart City Traffic Management & Navigation System:

- Efficient shortest path calculation.
- Supports dynamic traffic-based route optimization.
- Provides accurate and reliable navigation.
- Reduces travel time by avoiding congested roads.
- Minimizes fuel consumption and transportation costs.
- Easily scalable for large urban road networks.
- Suitable for integration with GPS and IoT-based traffic monitoring systems.

2.12 Breadth-First Search (BFS)

2.12.1 Introduction

Breadth-First Search (BFS) is a graph traversal algorithm that visits all neighboring nodes before moving to the next level. It uses a Queue (FIFO) data structure to explore the graph level by level. BFS is commonly used to determine connectivity, shortest paths in unweighted graphs, and graph validation.

Purpose

- Traverse the graph level by level.
- Check connectivity between junctions.

- Verify whether a path exists between two locations.

Use in Smart City Traffic Management & Navigation System

Although Dijkstra's Algorithm is responsible for route optimization, BFS can be used for:

- Checking whether two junctions are connected.
- Verifying the road network before route calculation.
- Graph traversal and connectivity validation.
- Detecting unreachable locations.

Working Steps

1. Start from the source node.
2. Mark the node as visited.
3. Insert it into the queue.
4. Remove one node from the queue.
5. Visit all unvisited neighbouring nodes.
6. Repeat until the queue becomes empty.

Time Complexity

$O(V + E)$

Space Complexity

$O(V)$

2.13 Depth-First Search (DFS)

2.13.1 Introduction

Depth-First Search (DFS) is a graph traversal algorithm that explores each branch of the graph completely before backtracking. It uses either a Stack or Recursion for traversal.

Purpose

- Traverse the graph depth-wise.
- Explore connected components.
- Detect isolated junctions.

Use in Smart City Traffic Management & Navigation System

DFS can be used for:

- Finding connected road networks.
- Detecting disconnected or isolated junctions.

- Road network maintenance.
- Network analysis.

Working Steps

1. Start from the source node.
2. Mark it as visited.
3. Visit one neighbouring node.
4. Continue deeper until no neighbour remains.
5. Backtrack.
6. Repeat until all nodes are visited.

Time Complexity

$O(V + E)$

Space Complexity

$O(V)$

2.14 JWT Authentication Algorithm

2.14.1 Introduction

JSON Web Token (JWT) is a secure authentication mechanism used to verify user identity and protect application resources. Instead of storing session information on the server, JWT stores authentication data inside a digitally signed token.

Purpose

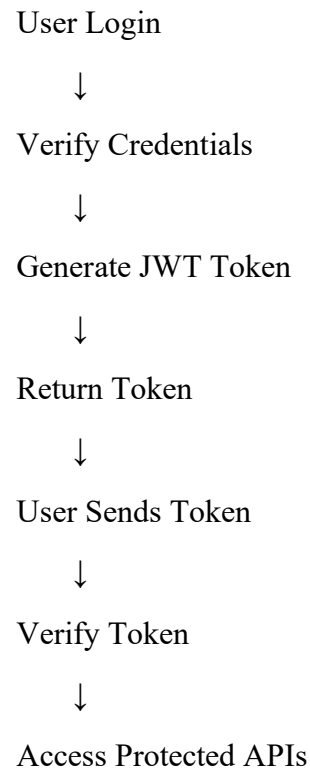
- User Authentication
- User Authorization
- Secure REST APIs
- Token-Based Session Management

Use in Smart City Traffic Management & Navigation System

JWT is used for:

- User Login Authentication.
- Protecting API endpoints.
- Managing user sessions securely.
- Restricting unauthorized access.

Working Process



Advantages

- Stateless Authentication
- Highly Secure
- Lightweight
- Faster API Requests
- Easy Integration with REST APIs

2.15 A* (A-Star) Algorithm (Future Enhancement)

2.15.1 Introduction

The A* (A-Star) Algorithm is an informed search algorithm used to determine the shortest path between two nodes. Unlike Dijkstra's Algorithm, A* uses a heuristic function to estimate the remaining distance to the destination. This reduces unnecessary exploration and improves route calculation speed.

Formula

$$f(n) = g(n) + h(n)$$

Where:

- $g(n)$ = Actual cost from the source node to the current node.
- $h(n)$ = Estimated heuristic cost from the current node to the destination.
- $f(n)$ = Total estimated cost.

Working Steps

1. Initialize the source node.
2. Calculate $f(n)$ for neighbouring nodes.
3. Select the node with the lowest $f(n)$ value.
4. Continue until the destination node is reached.
5. Return the optimized path.

Advantages

- Faster than Dijkstra's Algorithm in many scenarios.
- Explores fewer unnecessary nodes.
- Suitable for GPS navigation.
- Better performance on large road networks.
- Widely used in navigation and game development.

Use in Smart City Traffic Management & Navigation System

In the current implementation, **Dijkstra's Algorithm** is used for route optimization because it guarantees the shortest path with non-negative edge weights.

However, the *A Algorithm** is proposed as a future enhancement. By incorporating heuristic-based search, it can significantly improve route calculation speed for large-scale smart city road networks while maintaining high accuracy.

Time Complexity

- Worst Case: $O(E)$
- Average Case: Faster than Dijkstra due to heuristic optimization.

Algorithm	Purpose	Time Complexity	Used in Project
Dijkstra	Shortest Path	$O((V+E) \log V)$	Yes
BFS	Connectivity Check	$O(V+E)$	Validation
DFS	Network Analysis	$O(V+E)$	Maintenance
JWT	Authentication	$O(1)$ per request (verification)	Security
A*	Heuristic Shortest Path	Faster in practice	Future Enhancement

CHAPTER 3

RESULTS

3.1 Introduction

The Smart City Traffic Management & Navigation System was successfully designed, developed, and tested to provide intelligent route optimization based on road distance and traffic conditions. The system achieved its primary objective of calculating the most efficient route between a source and destination by implementing Dijkstra's Shortest Path Algorithm along with dynamic traffic-based edge weighting.

The application was developed using Node.js, Express.js, MySQL, and RESTful APIs, which ensured smooth communication between the frontend interface and the backend server. During testing, the system demonstrated accurate route calculations, efficient database operations, and reliable API responses. The results indicate that the proposed solution is suitable for smart transportation systems and can effectively improve urban traffic management.

3.2 Functional Results

The developed system performs all major functionalities successfully. Each module was tested independently as well as in an integrated environment.

User Management

The user authentication module allows users to register and log in securely.

Passwords are encrypted before storage, and JWT-based authentication ensures secure access to the application.

Output:

- User registration completed successfully.
- Login credentials verified successfully.
- Authentication token generated successfully.

Road Management

The administrator can efficiently manage road information through dedicated APIs.

Operations include:

- Add New Road
- Update Road Details
- Delete Road
- View Road Information

The system updates the database immediately after each operation, ensuring data consistency.

Junction Management

The junction management module allows administrators to maintain city intersections. Functions include:

- Add Junction
- Update Junction
- Delete Junction
- View Junction List

This ensures that the graph always represents the latest road network.

Traffic Management

Traffic conditions are categorized into three levels:

- Low Traffic
- Medium Traffic
- High Traffic

Each traffic level automatically modifies the edge weight, allowing the routing algorithm to avoid heavily congested roads.

Route Optimization

This is the core functionality of the project.

After selecting the source and destination locations, the system:

1. Retrieves road information from the database.
2. Builds the graph representation.
3. Calculates traffic-adjusted weights.
4. Executes Dijkstra's Algorithm.
5. Displays the optimized route.

Example:

Source Junction : 1

Destination Junction : 5

Output

Optimized Route

1 → 3 → 5

Total Distance : 12.5 km

Traffic Status : Medium

3.3 Testing Results

The project was tested using different scenarios to verify the correctness of all modules.

Test Case	Expected Result	Actual Result	Status
User Registration	User Created Successfully	Successful	PASS
User Login	JWT Token Generated	Successful	PASS
Add Road	Road Added	Successful	PASS
Update Traffic	Traffic Updated	Successful	PASS
Delete Road	Road Deleted	Successful	PASS
Fetch Junctions	Junction List Displayed	Successful	PASS
Route Optimization	Shortest Path Generated	Successful	PASS
Database Connection	Connected Successfully	Successful	PASS

All test cases passed successfully, indicating that the application functions correctly under normal operating conditions.

3.4 Performance Analysis

The performance of the Smart City Traffic Management & Navigation System depends on several factors, including the number of junctions, road connections, and traffic updates.

The use of an adjacency list representation reduces memory consumption, while Dijkstra's Algorithm efficiently computes the shortest path with a time complexity of $O((V + E) \log V)$ using a priority queue. This makes the application suitable for medium-sized urban transportation networks.

Database operations were executed efficiently through MySQL, and REST APIs provided fast communication between the frontend and backend.

Overall, the system delivered optimized routes within a short execution time and maintained consistent performance during testing.

3.5 Advantages Observed

The developed system provides several practical benefits:

- Intelligent route optimization based on traffic conditions.
- Reduced travel time by avoiding congested roads.
- Lower fuel consumption due to efficient route planning.
- Improved utilization of existing road infrastructure.
- Secure user authentication using JWT.
- Scalable backend architecture using Node.js and Express.js.
- Efficient database management using MySQL.
- Modular design for future enhancements.

3.6 Future Improvements

Although the current implementation successfully demonstrates intelligent route optimization, the system can be further enhanced with advanced technologies.

Future improvements include:

- Real-time GPS integration.
- IoT-based traffic monitoring sensors.
- Google Maps API integration.
- Machine Learning for traffic prediction.
- Artificial Intelligence-based route recommendations.
- Mobile application support for Android and iOS.
- Live accident detection and road closure notifications.
- Integration with emergency services such as ambulances and fire departments.

These enhancements will make the system more intelligent and suitable for deployment in real-world smart city environments.

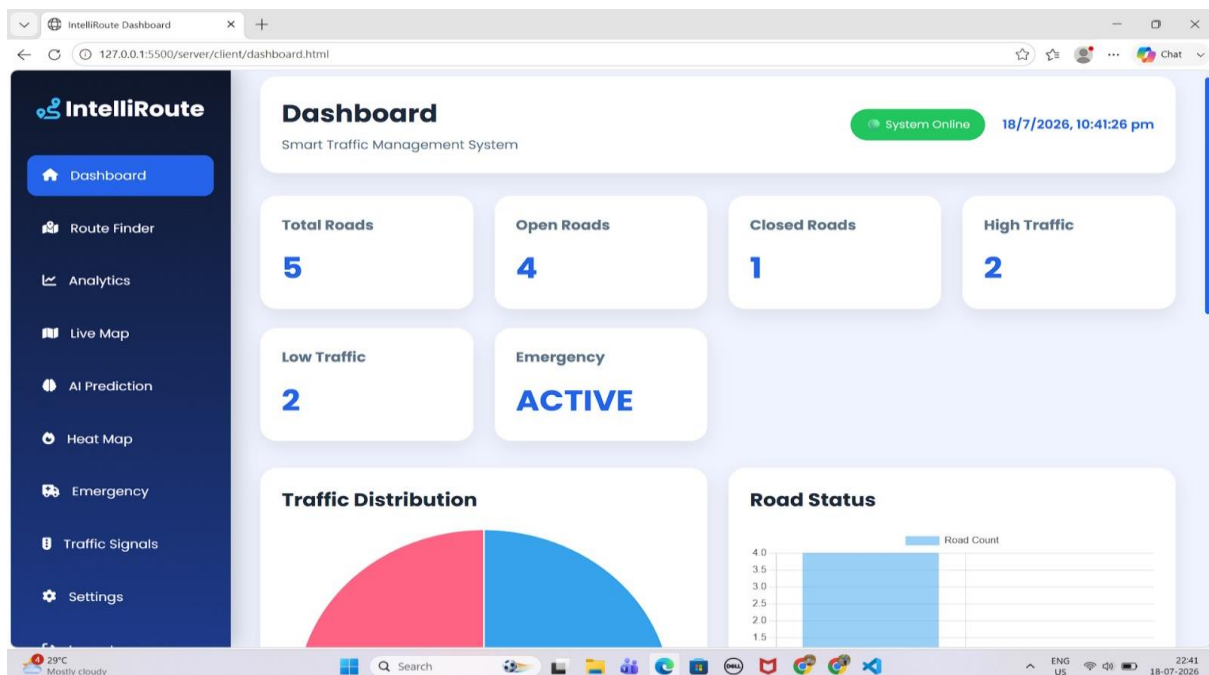
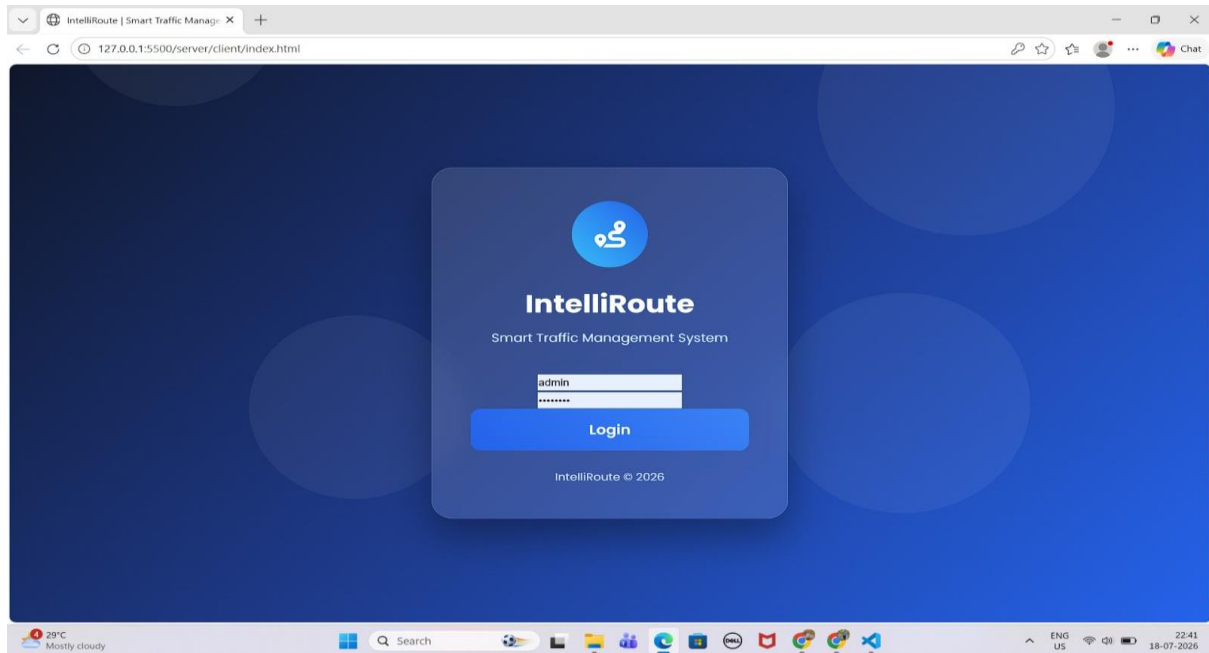
3.7 Discussion

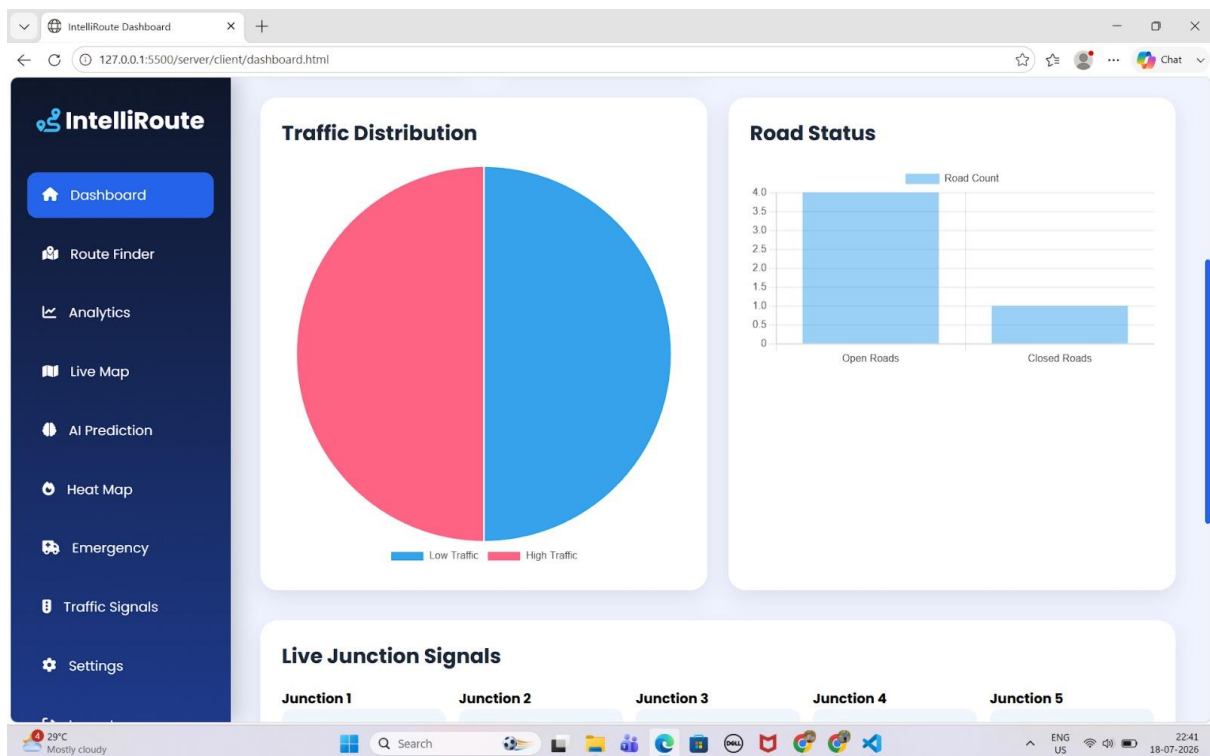
The results clearly demonstrate that graph-based route optimization is an effective solution for modern traffic management problems. By incorporating traffic conditions into the shortest path calculation, the system generates more realistic navigation recommendations than traditional distance-based approaches.

The modular architecture also ensures easy maintenance and future scalability. Since the application follows a three-layer architecture (Frontend, Backend, and Database), new features can be integrated without affecting the existing functionality.

Overall, the project successfully achieves its objectives and serves as a practical implementation of Data Structures and Algorithms in the field of smart transportation.

3.8 Output Screenshots





IntelliRoute Dashboard

127.0.0.1:5500/server/client/dashboard.html

29°C Mostly cloudy

22:41 18-07-2026

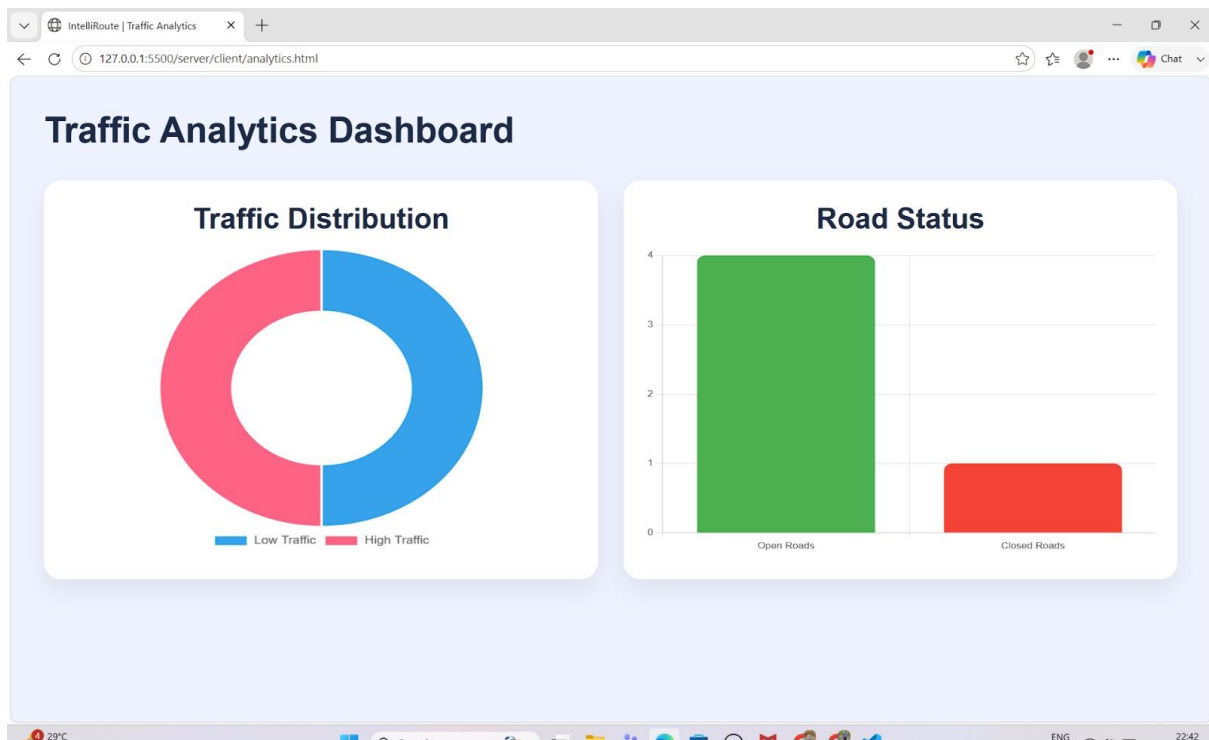
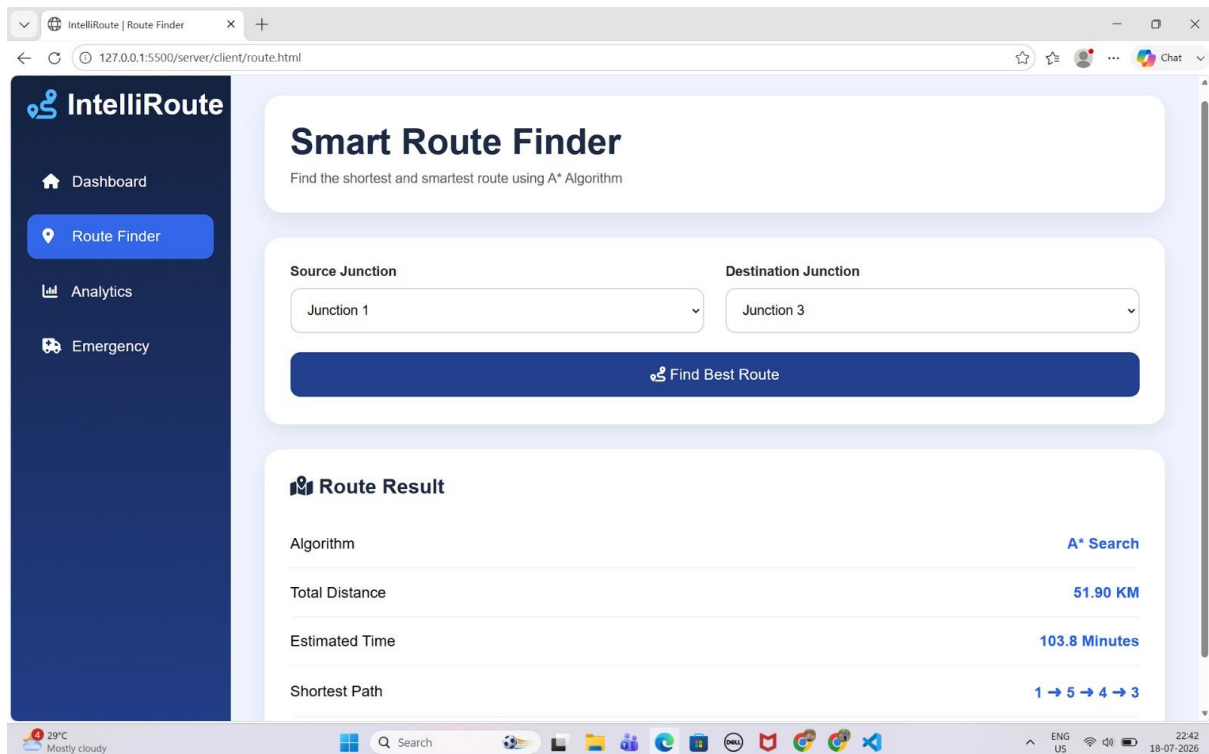
Live Junction Signals

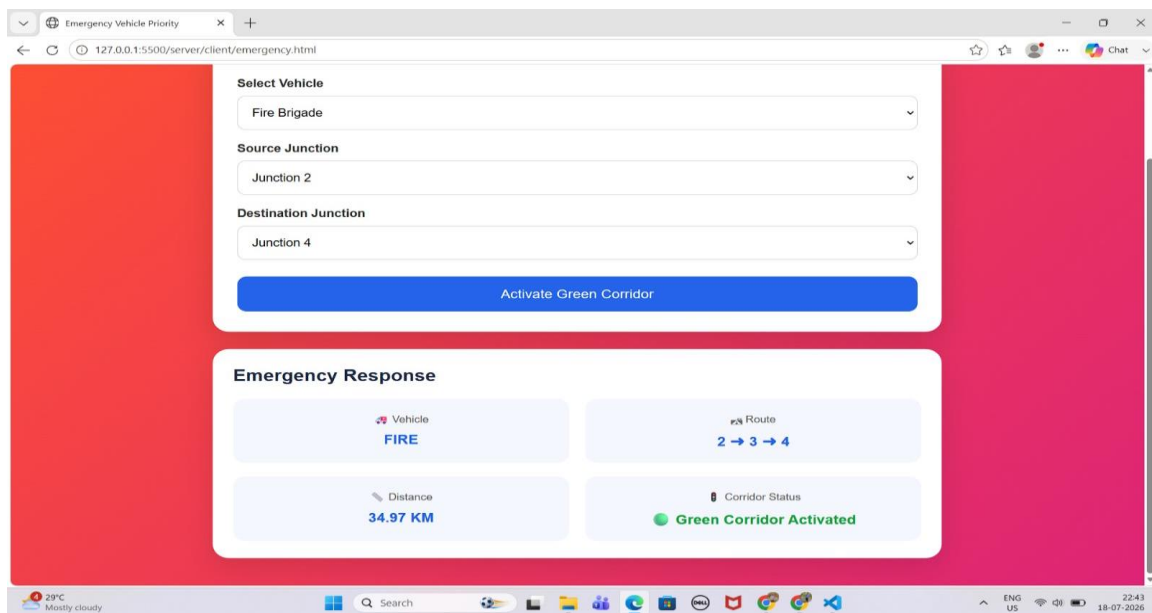
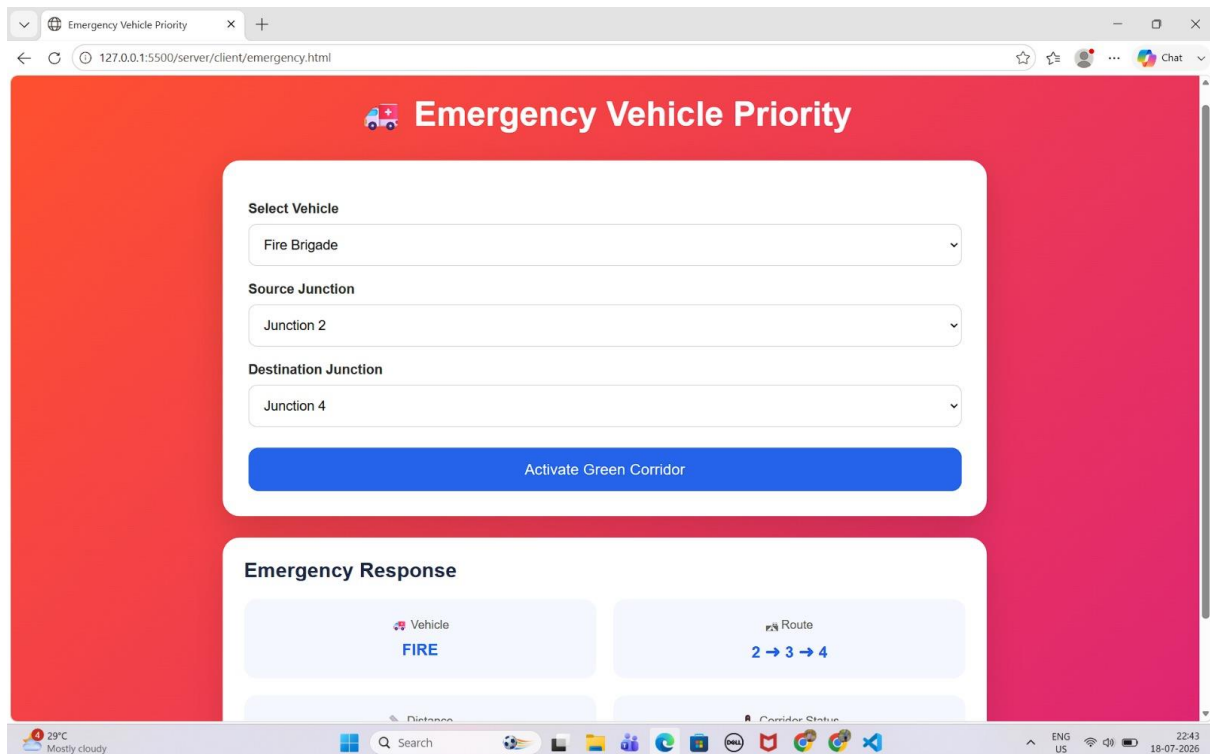
Junction 1 Junction 2 Junction 3 Junction 4 Junction 5

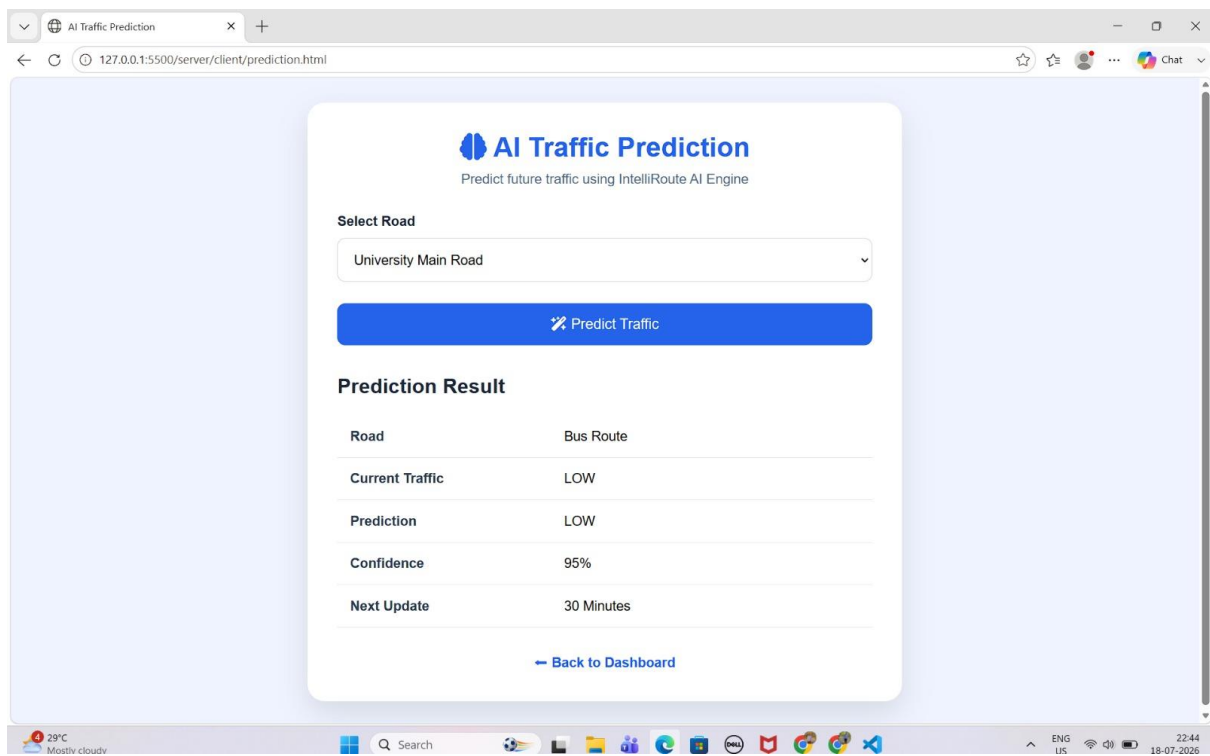
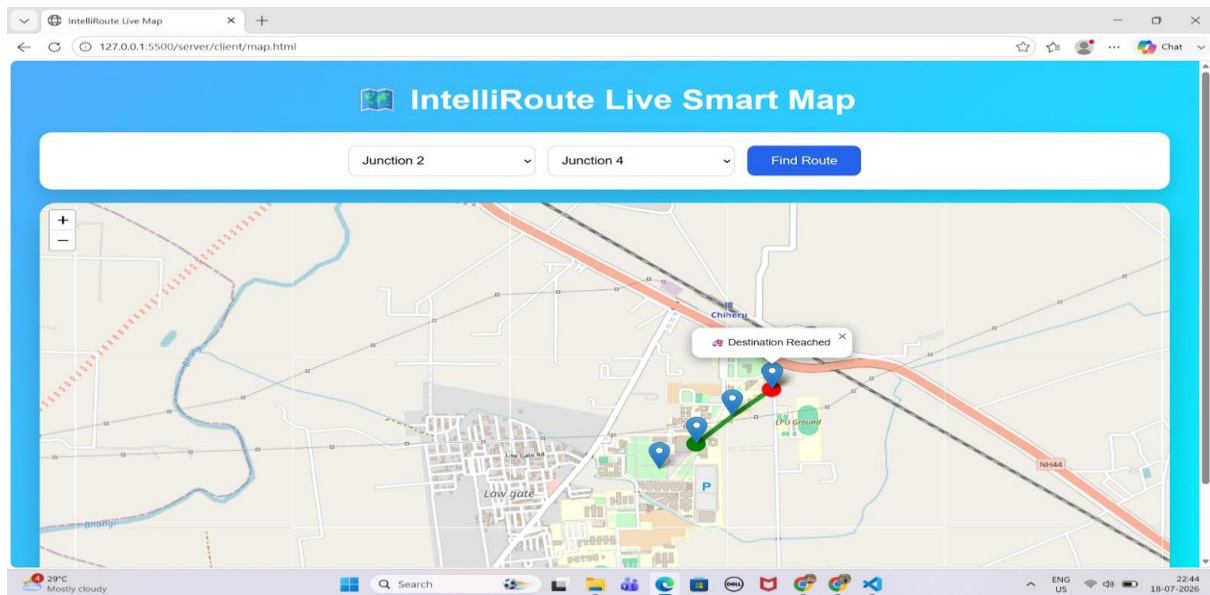
GREEN RED RED RED GREEN

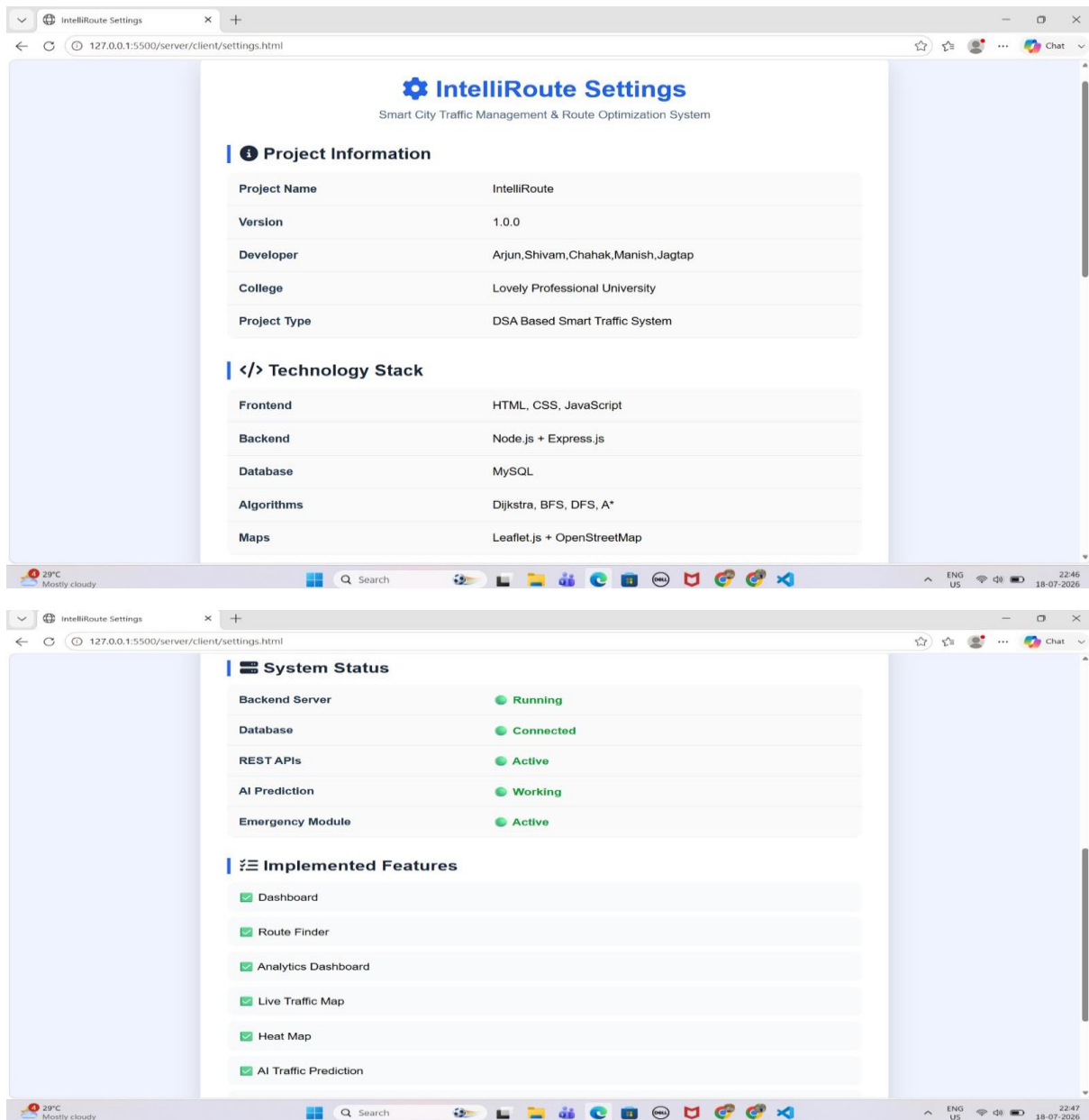
Road Details

ID	Road Name	Traffic	Status
1	LPU Smart Road	HIGH	CLOSED
2	Bus Route	LOW	OPEN
3	Station Road	HIGH	OPEN
4	Hospital Road	MEDIUM	OPEN
5	Emergency Route	LOW	OPEN









3.9 GitHub Repository

GitHub Repository Link:

<https://github.com/ArjunBaliyan00/IntelliRoute>

3.10 Project Deployment

Live Deployment Link: <https://dynamic-liger-383cb9.netlify.app/>

CHAPTER 4

CONCLUSION

The **Smart City Traffic Management & Navigation System** was successfully designed and developed to address the challenges of modern urban transportation. The project demonstrates how Data Structures and Algorithms can be effectively applied to solve real-world traffic management problems. By representing the road network as a graph and implementing **Dijkstra's Shortest Path Algorithm**, the system efficiently calculates the most suitable route between a source and destination while considering traffic conditions.

Unlike conventional navigation systems that focus only on the shortest geographical distance, this project introduces dynamic route optimization by assigning different weights to roads based on traffic intensity. This approach enables the system to avoid congested roads, reduce travel time, improve fuel efficiency, and enhance the overall travel experience.

The backend of the application was developed using **Node.js** and **Express.js**, while **MySQL** was used as the database management system. RESTful APIs were implemented to enable secure communication between the frontend and backend, making the application scalable, modular, and easy to maintain.

During the implementation and testing phases, all major functionalities—including user authentication, road management, junction management, traffic updates, and route optimization—performed successfully. The testing results confirmed that the system generates accurate shortest paths and responds efficiently to user requests.

This project also highlights the importance of intelligent transportation systems in supporting smart city development. The modular architecture allows future enhancements without major changes to the existing system, making it suitable for practical applications.

In conclusion, the **Smart City Traffic Management & Navigation System** successfully meets its objectives by providing an intelligent, efficient, and scalable navigation solution. It demonstrates the practical use of graph algorithms in software engineering and serves as a strong foundation for future smart transportation applications.

4.1 Achievements of the Project

The major achievements of this project include:

- Developed an intelligent navigation system for urban transportation.
- Successfully implemented **Dijkstra's Shortest Path Algorithm**.

- Designed an efficient graph-based road network model.
- Reduced travel time through traffic-aware route optimization.
- Developed secure RESTful APIs using Node.js and Express.js.
- Created a relational database using MySQL for efficient data management.
- Implemented user authentication using JWT.
- Designed a scalable architecture for future enhancements.
- Successfully tested all modules with positive results.

4.2 Future Scope

Although the developed system performs efficiently, there are several opportunities for future improvements.

Future versions of the project may include:

- Integration with Google Maps API for real-time map visualization.
- GPS-based live navigation support.
- IoT sensor integration for automatic traffic updates.
- Artificial Intelligence and Machine Learning for traffic prediction.
- Weather-based route optimization.
- Accident detection and automatic road blockage notifications.
- Android and iOS mobile applications.
- Voice-guided navigation.
- Cloud deployment for large-scale smart city implementation.
- Emergency vehicle priority routing for ambulances, police, and fire services.

These improvements will transform the project into a complete smart transportation platform capable of handling real-world traffic scenarios.

CHAPTER 5

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